

# Lecture VIII - Pandas and AI Craft

## Programming with Python

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### Checkpoint 4

**Sessions VI-VII.** The first 40 minutes are the checkpoint. It starts now, before the investor opens her data room.

- **Individual work:** no neighbors, no chat
- **AI tools are allowed:** being able to **VERIFY** the output is the skill being graded
- The **link and QR** are handed out in class. Open it and start
- ~6 short tasks: write code, fix a bug, a seeded draw, answer a multiple choice
- It sweeps **Sessions VI-VII:** modules, `random` and seeds, NumPy arrays and masks

### When you're done

Menu → *Download* → *Download Python code* → upload the `.py` to the “**Checkpoint 4**” assignment on Moodle. **No retakes:** one sitting.

...

#### Note

The green live checks are **provisional**. The final grading runs on my side. Everything in it was rehearsed in the labs.

## Episode 8: The Data Room

### Show me the data

Pens down. The checkpoint is behind you. This morning the rival chain MunchCorp put out a press release bragging about their “data-driven growth.” The evidence attached: exactly **one pie chart**. The investor slid it across the table and said, “*Yours will be better. Show me the data.*”

...

Tobi didn't wait. He'd pasted the question into a chatbot, proudly turned the laptop around, and hit run:

```
AttributeError: 'DataFrame' object has no attribute 'overview'
```

...

The AI sounded certain. The method it called **does not exist**. Today is about that difference, and about the tool that answers the investor for real: **pandas**.

## AI joins the team: professionally

### Prompting: context + constraints

Two things turn a vague request into a useful answer:

- **Context.** Say what the data is: “I have a DataFrame `orders` with columns `zone` (text) and `total_eur` (float).”
- **Constraints.** Say exactly what you want *back*: “Give me the total `total_eur` for zone `Nord`, as a single number.”

...

*“DataFrame `orders`, text column `zone`, float column `total_eur`. Write one line that returns the total `total_eur` where `zone` is `Nord`.”*

Vague in, vague out. Specific in, checkable out, and then you **iterate**.

### The verify workflow

Tobi’s mistake wasn’t *using* AI. It was **shipping without checking**. Every line an AI hands you gets three passes:

1. **Read it:** do you understand what each line claims to do?
2. **Run it:** does it actually execute, or does it crash?
3. **Test it:** try it on a small case where **you already know the answer**.

### Step 3 in action

Five orders with values you can add in your head, and the AI’s suggested line for the Nord average:

```
import pandas as pd

check = pd.DataFrame({
    "zone":      ["Nord", "Sued", "Nord", "Hafen", "Nord"],
    "total_eur": [10.0, 7.5, 20.0, 5.0, 30.0],
})
print(check[check["zone"] == "Nord"]["total_eur"].mean())
```

```
20.0
```

...

The Nord orders are 10, 20 and 30. You can average those in your head. Does the AI’s answer match? Then the line has earned some trust on eighty rows. Verification is the job now, not typing.

### Predict: it ran, so it's right?

Tobi asks the AI for the **Nord average** on the same five orders. It hands back one line and calls it done. What does this print?

```
print(check["total_eur"].mean())
```

a) 20.0 b) an `AttributeError` c) 14.5

...

Predict first. Pick a letter, then I reveal the answer.

### Answer: it ran, and it's wrong

c) **14.5**: the line averages **all five** orders. Nobody filtered for Nord, and pandas has no way to know that was the question:

```
print(check["total_eur"].mean())           # every zone
print(check[check["zone"] == "Nord"]["total_eur"].mean()) # Nord only
```

```
14.5
20.0
```

...

No traceback, a plausible number, and still wrong. "It runs" only clears step 2; step 3 is where the investor's number gets tested.

### Predict: does `.summarize()` exist?

The AI wrote this for Tobi. `orders` is a tiny two-row frame. What happens on the last line?

```
import pandas as pd
orders = pd.DataFrame({"zone": ["Nord", "Sued"], "total_eur": [12.0, 9.5]})

orders.summarize()
```

a) prints a summary table b) raises an `AttributeError` c) returns an empty DataFrame

...

Predict first. Pick a letter, then I reveal the answer.

### Answer: the method never existed

b) **`AttributeError`**: pandas has no `.summarize()`. The AI invented it:

```
import pandas as pd
orders = pd.DataFrame({"zone": ["Nord", "Sued"], "total_eur": [12.0, 9.5]})
orders.summarize()           # the method the AI invented
```

```
AttributeError: 'DataFrame' object has no attribute 'summarize'
[31mAttributeError[39m[31m: [39m 'DataFrame' object has no attribute
'summarize'
```

...

The real method is `.describe()`:

```
print(orders.describe())          # the method that actually exists
```

```
      total_eur
count    2.000000
mean    10.750000
std      1.767767
min      9.500000
25%     10.125000
50%     10.750000
75%     11.375000
max     12.000000
```

...

**An AI that sounds sure is not the same as an API that exists.**

## Disclosure: one line, every time

From Session VI the course rule stands: every submission that used AI carries a **one-line note** saying what you used it for.

- “Used the chatbot to draft the `groupby` line; I checked the totals by hand.”
- Standard practice, not an admission of anything.

...

It protects **you**: it separates the work you understand from the work you pasted, so when a reviewer (or the investor) asks “how does this line work?”, you’re never caught claiming something you can’t explain.

## When NOT to reach for AI

Sometimes the fastest path is the one you already own. You built a whole muscle in Part I:

- A red traceback? **You’ve read those since Episode 5.** The last line names the error and points at the file, often faster than describing it to a chatbot.
- A one-line filter you’ve written ten times? Just write it.

...



Tip

Use AI to draft the unfamiliar and to explain the confusing, not to skip thinking you can do yourself.

## Your turn: 5-10 minutes

### Open the exercise

Scan the QR or type the link:

[python.tobiasvlcek.com/notebooks/ex\\_08\\_a/](https://python.tobiasvlcek.com/notebooks/ex_08_a/)



First **predict** what happens, then run it.

## The DataFrame

### What pandas adds over NumPy

Last session, a NumPy array turned a thousand numbers into one fast object, but every value had to share **one type**, and columns had no names.

- A **DataFrame** is a table: named columns, and each column can be a different type
- Text zones next to float euros next to integer counts, all in one object
- It's the spreadsheet the investor handed you, loaded into Python

...

The convention everyone uses: `import pandas as pd`.

### A DataFrame from a dictionary

Keys become **column names**; each list becomes a **column**; one order per row, mixed types together:

```
import pandas as pd
df = pd.DataFrame({
    "order_id": [101, 102, 103, 104, 105],
    "zone":     ["Nord", "Sued", "Nord", "Hafen", "Sued"],
    "items":    [2, 1, 3, 1, 4],
    "total_eur": [18.50, 7.20, 24.00, 6.80, 31.40],
})
print(df)
```

|   | order_id | zone  | items | total_eur |
|---|----------|-------|-------|-----------|
| 0 | 101      | Nord  | 2     | 18.5      |
| 1 | 102      | Sued  | 1     | 7.2       |
| 2 | 103      | Nord  | 3     | 24.0      |
| 3 | 104      | Hafen | 1     | 6.8       |
| 4 | 105      | Sued  | 4     | 31.4      |

...

#### Note

Nobody types eighty orders by hand: today's lab loads a real CSV in one line, `orders = pd.read_csv("public/orders.csv")`. In the browser that line fetches over the web instead of from disk; your code doesn't change.

## First look: `.head()` and `.info()`

Before analyzing a table, glance at it. `.head()` shows the top rows; `.info()` is your first sanity check: right number of rows? Any column a surprising type?

```
print(df.head(3))    # first 3 rows
df.info()            # columns, dtypes, non-null counts
```

|   | order_id | zone | items | total_eur |
|---|----------|------|-------|-----------|
| 0 | 101      | Nord | 2     | 18.5      |
| 1 | 102      | Sued | 1     | 7.2       |
| 2 | 103      | Nord | 3     | 24.0      |

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   order_id    5 non-null      int64
1   zone        5 non-null      object
2   items       5 non-null      int64
3   total_eur   5 non-null      float64
```

```
dtypes: float64(1), int64(2), object(1)
memory usage: 292.0+ bytes
```

## The numbers at a glance: `.describe()`

`.describe()` computes count, mean, min, max and quartiles for every **numeric** column at once:

```
print(df.describe())
```

|       | order_id   | items   | total_eur |
|-------|------------|---------|-----------|
| count | 5.000000   | 5.00000 | 5.000000  |
| mean  | 103.000000 | 2.20000 | 17.580000 |
| std   | 1.581139   | 1.30384 | 10.688873 |
| min   | 101.000000 | 1.00000 | 6.800000  |
| 25%   | 102.000000 | 1.00000 | 7.200000  |
| 50%   | 103.000000 | 2.00000 | 18.500000 |
| 75%   | 104.000000 | 3.00000 | 24.000000 |
| max   | 105.000000 | 4.00000 | 31.400000 |

...

One call, the whole numeric summary: this is the `.summarize()` the AI wished existed, spelled correctly.

## Selecting a column, filtering rows

Pick one column by name; keep rows with a **boolean mask**, exactly the NumPy idea from last session:

```
print(df["zone"])          # one column (a Series)
print()
print(df[df["zone"] == "Nord"])  # only the Nord rows, case-sensitive!
```

```
0    Nord
1    Sued
2    Nord
3    Hafen
4    Sued
Name: zone, dtype: object
```

|   | order_id | zone | items | total_eur |
|---|----------|------|-------|-----------|
| 0 | 101      | Nord | 2     | 18.5      |
| 2 | 103      | Nord | 3     | 24.0      |

...

`df["zone"] == "Nord"` builds a column of `True/False`; indexing with it keeps the `True` rows. `.loc` exists for label-based selection, but a plain mask covers today.

## Predict: what does a comparison return?

Tobi wants the orders over 20 EUR. He compares the column and prints the result:

```
big = df["total_eur"] > 20
print(big)
```

a) a column of `True/False`, one per row b) only the rows whose total is over 20 c) a single `True`, since some order is over 20

...

Predict first. Pick a letter, then I reveal the answer.

## Answer: a mask, not a table

a) a column of `True/False`. The comparison alone is the **mask**, a Series with one boolean per row. The rows only appear once you index with it:

```
big = df["total_eur"] > 20
print(big)                # the mask: a Series of booleans
print(df[big])            # the rows: build the mask, then apply it
```

```
0    False
1    False
2     True
3    False
4     True
Name: total_eur, dtype: bool
   order_id  zone  items  total_eur
2        103  Nord     3        24.0
4        105  Sued     4        31.4
```

## Your turn: 5-10 minutes

### Open the exercise

Scan the QR or type the link:

[python.tobiasvlcek.com/notebooks/ex\\_08\\_b/](https://python.tobiasvlcek.com/notebooks/ex_08_b/)



First **predict** what happens, then run it.

### Adding a column, sorting

Compute a new column from existing ones, then sort. Work on a **copy** so the original stays intact:

```
priced = df.copy()
priced["eur_per_item"] = priced["total_eur"] / priced["items"]
print(priced.sort_values("total_eur", ascending=False))
```

|   | order_id | zone  | items | total_eur | eur_per_item |
|---|----------|-------|-------|-----------|--------------|
| 4 | 105      | Sued  | 4     | 31.4      | 7.85         |
| 2 | 103      | Nord  | 3     | 24.0      | 8.00         |
| 0 | 101      | Nord  | 2     | 18.5      | 9.25         |
| 1 | 102      | Sued  | 1     | 7.2       | 7.20         |
| 3 | 104      | Hafen | 1     | 6.8       | 6.80         |

...

`sort_values` reorders rows by a column, biggest orders first here. The new column is just arithmetic on two others, computed for every row at once.

### Predict: filtering with the wrong case

Tobi filters for the Nord zone, but types it **lowercase**. The data spells it "Nord". What does this print?

```
print(df[df["zone"] == "nord"])
```

a) an empty table, no error b) the Nord rows anyway: case is ignored c) a `KeyError` for the missing zone

...

Predict first. Pick a letter, then I reveal the answer.

### Answer: an empty table, no warning

a) an empty table. "nord" matches nothing, so the mask is all `False` and pandas hands back **zero rows**. No error, no complaint:

```
print(df[df["zone"] == "nord"])          # nothing matches "nord"
print("rows:", len(df[df["zone"] == "nord"]))
```

```
Empty DataFrame
Columns: [order_id, zone, items, total_eur]
Index: []
rows: 0
```

...

This is the dangerous kind of bug: it **runs**, it just returns nothing. **pandas won't warn you; YOU verify**, which is exactly why you test filters on a case whose answer you know.

### One more tool: `groupby`

The investor's real question is **per zone**: which area brings in the most?

- `groupby` splits the table by a category, then computes **once per group**
- `df.groupby("zone")["total_eur"].sum()` → one total for each zone

...

#### Note

That's the teaser: today's lab does the heavy lifting with `groupby`, turning eighty raw orders into the handful of numbers the investor actually asked for.

## Your turn: 5-10 minutes

### Open the exercise

Scan the QR or type the link:

[python.tobiasvlcek.com/notebooks/ex\\_08\\_c/](https://python.tobiasvlcek.com/notebooks/ex_08_c/)



First **predict** what happens, then run it.

## To the Lab

### After the break: the lab

- Head to the lab notebook: [Episode 8: The Data Room](#), in your browser as always
- The investor slides a **USB stick** across the table: every order of two weeks, now `orders.csv`, eighty rows, one `pd.read_csv` line
- You'll `.head()` it, ask its `.shape`, filter with masks, add a column on a safe copy, and answer per-zone questions with `groupby`
- Tobi has discovered AI; your real job is to **supervise** it and catch the confident nonsense

...

#### ! Important

**Download your `.py` before you leave.** Closing the tab without downloading loses your work, the same motion you used to hand in the checkpoint at the start of the session.

## Wrap-up

### Three things to remember

1. **AI drafts, you verify.** Give it context and constraints, then read it → run it → test it on a case you know.
2. **A DataFrame is a named, mixed-type table.** `pd.DataFrame` from a dict, `pd.read_csv` from a file; `.head()`, `.info()`, `.describe()` to look before you leap.

3. **Select, filter, add, group.** `df["col"]`, a boolean mask (`df[df["zone"] == "Nord"]`, case-sensitive!), a new column on a `.copy()`, and `groupby` for per-category answers. pandas won't warn you; you verify.

...

#### Note

**Next episode: charts the investor can't argue with.** Numbers convince the careful; a good plot convinces the room. We turn the data room into pictures.

## Literature

### Books to start with

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. [Link to free online version](#)
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

...

#### Note

Working with AI this session? Revisit the [AI Tools page](#): context-and-constraints prompting, the verify workflow, and the one-line disclosure habit. For pandas itself, the official [10 minutes to pandas](#) guide is the friendliest next step.

...

For more, see the [literature list](#) of this course.